

TEST CODE 01238010

JANUARY 2018

FORM TP 2018021

CARIBBEAN EXAMINATIONS COUNCIL
CARIBBEAN SECONDARY EDUCATION CERTIFICATE®
EXAMINATION
PHYSICS

Paper 01 – General Proficiency

1 hour 15 minutes

17 JANUARY 2018 (a.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

This test consists of 60 items. You will have 1 hour and 15 minutes to answer them.

In addition to this test booklet, you should have an answer sheet.

Each item in this test has four suggested answers lettered (A), (B), (C), (D). Read each item you are about to answer and decide which choice is best.

On your answer sheet, find the number which corresponds to your item and shade the space having the same letter as the answer you have chosen. Look at the sample item below.

Sample Item

The SI unit of length is the

- (A) metre
- (B) second
- (C) newton
- (D) kilogram

Sample Answer

☒ (A) ☐ (B) ☐ (C) ☐ (D)

The best answer to this item is “metre”, so (A) has been shaded.

If you want to change your answer, erase it completely before you fill in your new choice.

When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot answer an item, go on to the next one. You may return to that item later.

You may do any rough work in this booklet.

Figures are not necessarily drawn to scale.

You may use a silent, non-programmable calculator to answer items.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

1. When used in front of a unit the prefix 'mega' means

(A) 10^{-6}
(B) 10^{-3}
(C) 10^3
(D) 10^6

D-
Mega- $\times 10^6$

2. 0.0000462 N expressed in standard form is

(A) 4.62×10^{-5} N
(B) 4.62×10^4 N
(C) 4.62×10^{-4} N
(D) 4.62×10^5 N

A

3. One gram is equal to

(A) 10 milligrams
(B) 100 milligrams
(C) 1 000 milligrams
(D) 10 000 milligrams

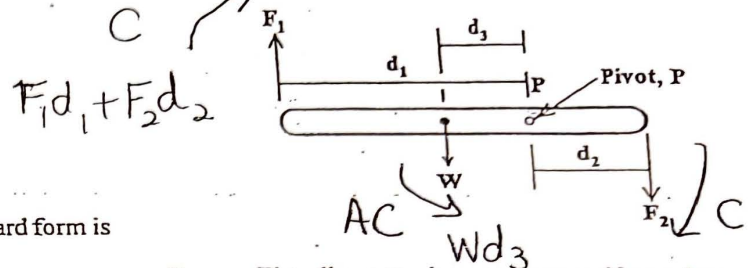
C

4. When a student placed her hands close to a television screen, the hair on her hands stood up pointing towards the screen. This is an example of

(A) a nuclear force
(B) an electric force
(C) a magnetic force
(D) a frictional force

B

Item 5 refers to the following diagram.



5. The diagram shows a non-uniform bar pivoted at P. W indicates the weight of the bar and forces F_1 and F_2 are applied at the positions shown.

The sum of the clockwise moments about P equals

(A) $F_2 d_2$
(B) $F_1 d_1 + W d_3$
(C) $(F_1 + W)(d_1 + d_2)$
(D) $F_1 d_1 + F_2 d_2$

D

6. Which of the following are examples of alternative sources of energy used in the Caribbean?

I. Nuclear energy
II. Geothermal energy
III. Hydroelectric energy

(A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

C

Pressure increases with depth

-3-

7. A diving ball is to be used at great depths and so its walls are made thick. The MAIN reason for this is that

(A) water pressure increases with depth
(B) the water is much colder at greater depths
(C) a thin-walled vessel would float up to the surface
(D) the density of water is much greater at great depths

A

8. An object of mass, m , is attached to a spring balance and its weight, w , recorded. What will be the result if the object is taken to the moon where the gravitational field strength is less, and weighed?

(A) Mass = m ; weight less than w
(B) Mass less than m ; weight = w
(C) Mass = m ; weight greater than w
(D) Mass greater than m ; weight = w

A
g on the moon is less than earth

9. A hydroelectric power station uses a renewable source of energy, X. This energy raises water to the top of a mountain so that it has gravitational potential energy. As the water runs down the mountain, it turns a turbine which converts Y energy into Z energy. Which set of answers for X, Y and Z is correct?

	X	Y	Z
(A)	Electrical	Potential	Kinetic
(B)	Geothermal	Potential	Electrical
(C)	Solar	Kinetic	Electrical
(D)	Chemical	Kinetic	Electrical

Renewable → solar
Motion of Turbine → kinetic
Product of generator → Electrical

M X Y

W Y

Y less dense
X as it float

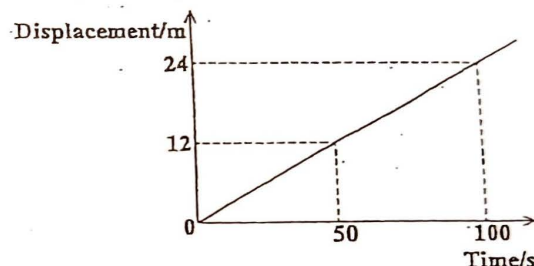
10. A solid object, X, floats in mercury and sinks in water. A solid object, Y, floats in both mercury and water.

Which of the following is true about X and Y?

(A) X is less dense than Y.
(B) X is more dense than Y.
(C) X and Y are both denser than water.
(D) X and Y are both denser than mercury.

B

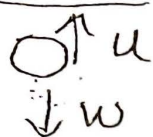
Item 11 refers to the following graph.



11. The graph above shows how the displacement of a runner from a starting line varies with time. This runner is

(A) not moving
(B) moving faster and faster
(C) moving at a steady speed
(D) moving slower and slower

C - gradient = velocity is constant



$u > w$ hence rises

- 4 -

B

12. A bubble of gas rises to the surface of a soft drink. This is because the

- (A) density of the gas is greater than the density of the drink
- (B) upthrust on the bubble is greater than the weight of the bubble
- (C) upthrust on the bubble is greater than the weight of water it displaces
- (D) weight of the water displaced by the bubble is less than the weight of the bubble

13. A block is allowed to fall freely towards the ground. As it falls, its gravitational potential energy

- (A) increases
- (B) remains constant
- (C) is converted to internal energy
- (D) is converted to kinetic energy

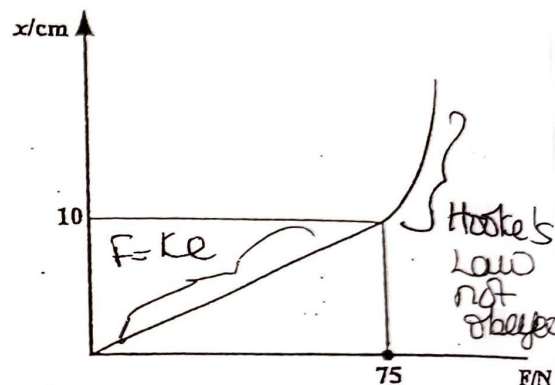
14. Pressure in a liquid can be calculated using the formula $P = \rho gh$.

Which of the following sets of units will give the pressure in the SI unit?

	ρ	g	h
(A)	g cm^{-3}	m s^{-2}	mm
(B)	kg m^{-3}	N kg^{-1}	m
(C)	g cm^{-3}	N kg^{-1}	m
(D)	kg m^{-3}	cm s^{-2}	cm

B

Item 15 refers to the following graph of a light spring which shows a simple extension, x , versus force, F .



15. Which of the following statements are true?

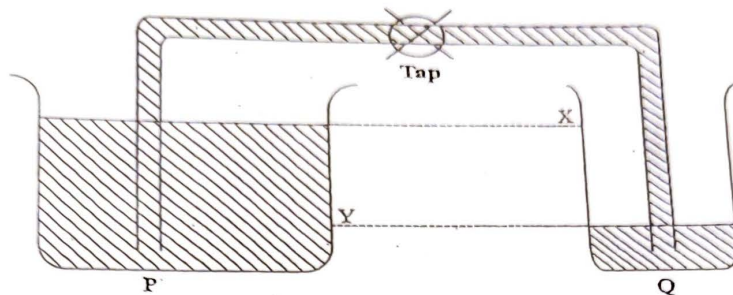
- I. The elastic limit of the spring was exceeded.
- II. The spring obeyed Hooke's law over its entire extension.
- III. The force per unit extension in the elastic region was 7.5 N cm^{-1} .

B

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

Diagram of a block falling with a downward arrow. Handwritten notes: "vel inc as it falls", "GPE $\rightarrow$ KE".

Item 16 refers to the following diagram which shows two containers of different sizes, with water at different levels connected by a glass tube and controlled by a tap.



16. When the tap is opened, water will flow from P to Q until

- (A) container P is empty
(B) the water level of Q is at X
(C) the water level of P is at Y
(D) the water level of P and Q are equal

water will flow until
the pressure is equalized
This occurs at the same
depth.

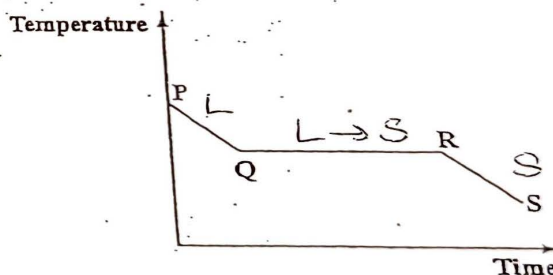
17. Two smooth spheres, A and B, collide head on. Which of the following statements is/are true?

- I. The momentum of A is the same after collision as it was before.
II. The momentum of B is the same after collision as it was before.
III. The total momentum of A and B is the same after collision as it was before.

- (A) I only
(B) III only
(C) II and III only
(D) I, II and III

With certainty
only the principle
of conservation
of momentum
is valid

Item 18 refers to the following graph based on an experiment on change of phase.

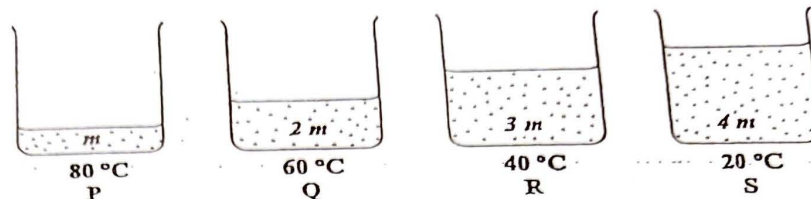


18. The graph shows that solidification started at Q. During which of the stages is the substance in the liquid phase?

- (A) At P only
(B) Between Q and R
(C) Between R and S
(D) Between P and Q

$$\frac{3}{2} kT$$

Item 19 refers to the following diagrams where P, Q, R and S are identical containers containing water of masses m , $2m$, $3m$ and $4m$ respectively at the temperatures indicated.



19. In which of the containers above is the average kinetic energy of the molecules GREATEST?

A

(A) P
(B) Q
(C) R
(D) S

Avg KE = $\frac{1}{2}mv^2$
greatest vel at
highest T

20. The specific heat capacity of a substance is defined as the quantity of thermal energy

- D
- (A) the substance can hold
(B) 1 kg of the substance can hold
(C) required to raise the temperature of the substance by 1 degree Celsius
(D) required to raise the temperature of 1 kg of the substance by 1 degree Celsius

22. Which of the following is NOT an example of evaporation? L → G

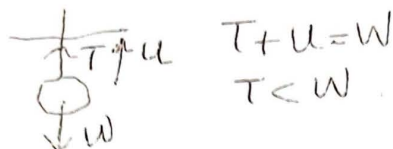
- (A) The cooling effect of sweating in animals
(B) A slice of bread left in the open air becomes dry
(C) The rapid disappearance of ether if exposed to the air
(D) A copper wire put around a block of ice gradually cuts through the ice

21. In the pressure law, which of the following statements is true?

- B
- (A) Pressure is constant.
(B) Volume is constant.
(C) The ratio of volume to pressure is constant.
(D) The ratio of volume to temperature is constant.

23. Who was responsible for arriving at the conclusion that measured amounts of electrical and mechanical energy can be converted to proportionate amounts of heat energy?

- A
- (A) Joule → Paddle exp t
(B) Newton
(C) Rumford
(D) Coulomb



$$E_h = m l_f \quad l_f = \text{const}$$

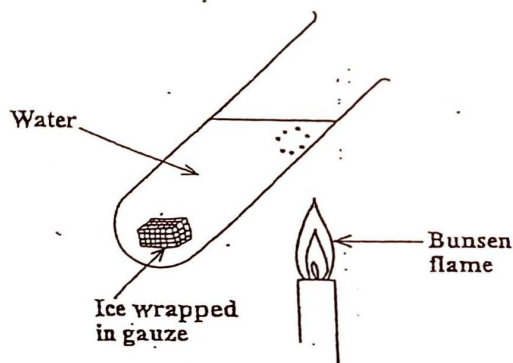
$$-7- \quad 2E_h = (2m) l_f$$

24. A piece of string is tied onto a small stone and the stone is then totally immersed and suspended in water. The tension in the string will be

(A) zero
(B) equal to the weight of the stone
(C) less than the weight of the stone
(D) more than the weight of the stone

C

Item 25 refers to the following diagram which shows water boiling at the top of a glass test tube while a piece of ice remains unmelted at the bottom.



25. Which of the following statements provides the reason for this occurrence?

(A) Gauze is a poor conductor of heat.
(B) Water is a good conductor of heat.
(C) Water is a poor conductor of heat.
(D) Glass is a good conductor of heat.

C

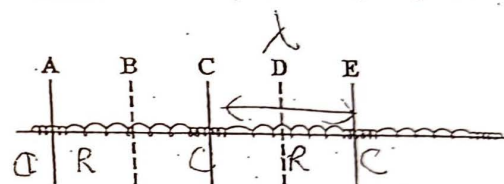
Water will not
conduct heat
to melt ice

26. The energy required to change the state of a substance was determined to be E_h . If the mass of the substance was DOUBLED, the value of E_h will be

(A) doubled
(B) quadrupled
(C) halved
(D) constant

A

Item 27 refers to the following diagram.



27. The diagram above illustrates a longitudinal wave train produced on a slinky. Which of the following distances represent the wavelength of the vibration?

(A) A to B
(B) A to E
(C) C to E
(D) D to E

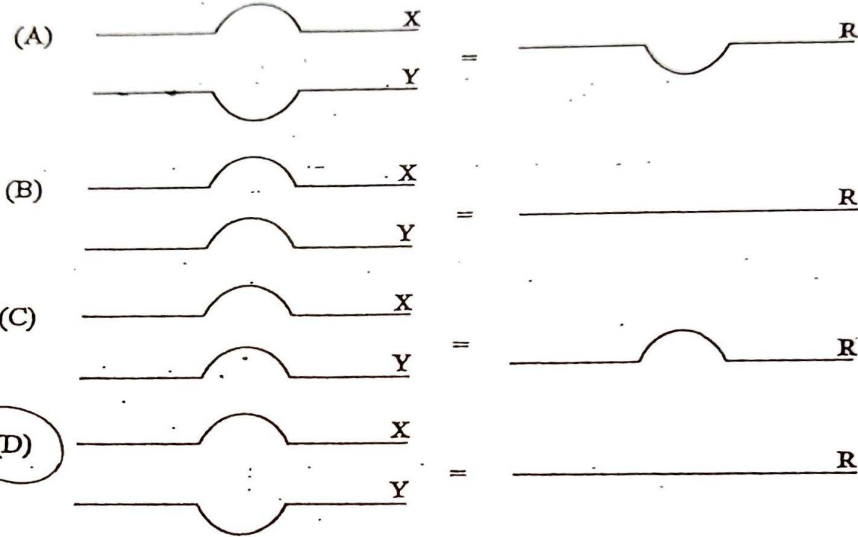
C

28. The phenomenon in which a wave bends as it passes the edge of an object is known as

(A) refraction
(B) diffraction
(C) rarefraction
(D) interference

B

29. In the following diagrams, X and Y represent two wave pulses superposing on each other to give a resultant R. Which diagram BEST illustrates the principle of superposition?



Crest +
Trough =
cancellation
of wave.

30. Two types of radiation, L and R, fall on a woman's left and right hands respectively. Her left hand feels hot but does not become suntanned; her right hand does not feel hot but it eventually becomes suntanned.

Radiations L and R are, respectively,

	L	R
(A)	visible light	X-rays
(B)	X-rays	visible light
(C)	ultraviolet	infrared
(D)	infrared	ultraviolet

C

31. A sand wave of frequency, f , moves with a speed, v . The wavelength of this wave is given by

- (A) fv
(B) $\frac{f}{v}$
(C) $\frac{v}{f}$
(D) $\frac{1}{f}$



Infrared -
emitted
by
hot objects

UV →
used
specifically
for sun tan.

32. Which of the following are laws of refraction?

- I. $\frac{\sin i}{\sin r} = \text{constant}$
 II. Angle of incidence = angle of refraction
 III. The incident ray, the refracted ray and the normal at the point of incidence all lie in the same plane

- (A) I and II only
 (B) I and III only
 (C) II and III only
 (D) I, II and III

33. A ray of light leaves air and enters glass of refractive index 1.6. The angle of refraction is 27° . What is the sine of the angle of incidence?

- (A) $1.6 + \sin 27^\circ$
 (B) $\frac{1.6}{\sin 27^\circ}$
 (C) $1.6 \sin 27^\circ$
 (D) $\frac{\sin 27^\circ}{1.6}$

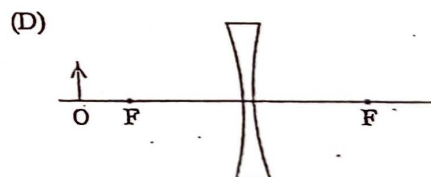
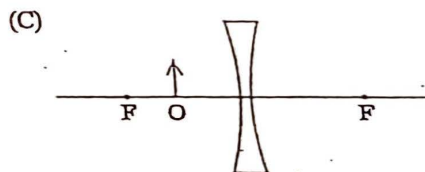
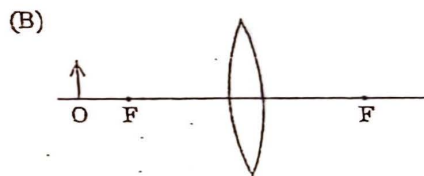
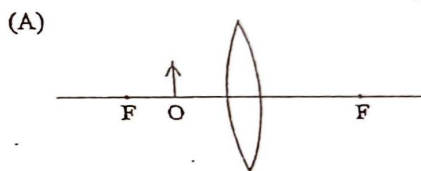
$$n_a \sin i = n_g \sin r = \frac{n_g \sin \theta_g}{n_a \sin \theta_g}$$

$$1.6 = \frac{\sin i}{\sin 27^\circ}$$

$$\sin i = 1.6 \times \sin 27^\circ$$

34. In each of the following diagrams, F represents the position of the principal focus, and O the position of the object for the given lens. Which arrangement will produce a real image?

$u > f$ for a convex lens



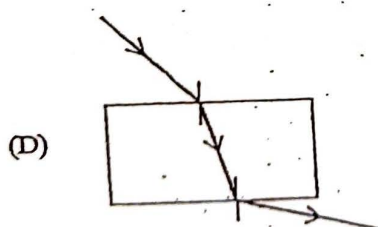
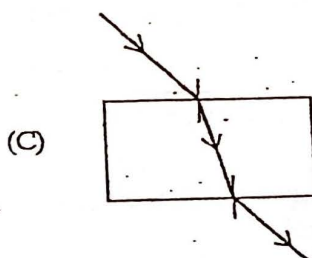
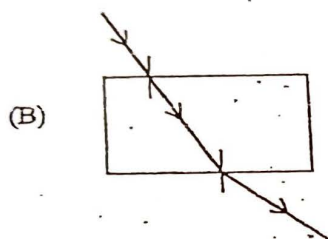
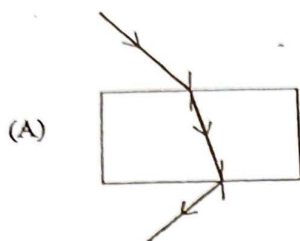
Concave lens $\Rightarrow$ virtual smaller & upright

35. Which of the following waves are NOT transverse waves?

- (A) A ripple on a pond
 (B) Light waves from a street lamp
 (C) Sound waves from a guitar string
 (D) Radio waves travelling from a satellite

Sound $\rightarrow$ Long

36. Which diagram BEST shows the path taken by a ray of light through a rectangular block?



C
angle of
 i = angle of
emergent
ray.

37. Which of the following combinations of ammeter characteristics is correct?

	Resistance of an ammeter	How is it connected to the component?
(A)	Low	In series
(B)	High	In series
(C)	Low	In parallel
(D)	High	In parallel

A

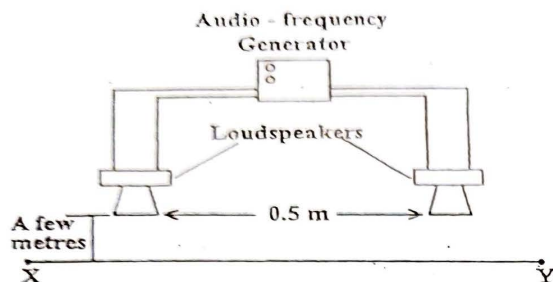
38. In an experiment, the approximate method is used to estimate the focal length of a converging lens. Why is it advised to use a distant object?

- (A) Rays can be seen better.
 (B) Rays are almost parallel.
 (C) Rays travel in straight lines.
 (D) Rays diverge better at the lens.

B

Rays emerging from a distant object tend to be parallel

Item 39 refers to the following diagram.



39. Two similar loudspeakers are connected to the same audio-frequency generator and set up as shown. At some points along XY no sound is heard because.

- (A) the sound waves are diffracted
(B) sound is refracted away from those points
(C) interference of the sound waves takes place *Destructive*
(D) the sound waves are reflected back to the same source

40. The human ear is incapable of hearing a silent dog whistle because.

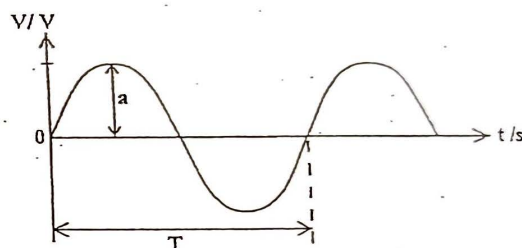
- (A) it does not make a noise
(B) the speed of sound is too fast to be detected by the human ear
(C) the frequency of the waves are above the range detected by the human ear
(D) the waves are infrasound with frequencies below that detectable by the human ear

Range of human hearing: 20 Hz - 20000 Hz

41. When a polythene rod is rubbed with a cloth, it becomes

- (A) positively charged by gaining protons
(B) negatively charged by gaining electrons
(C) positively charged by gaining electrons
(D) negatively charged by losing protons

Item 42 refers to the following voltage-time graph.



42. The frequency of the voltage-time graph is given by

- (A) a
(B) $\frac{1}{a}$
(C) T
(D) $\frac{1}{T}$

$$f = \frac{1}{T}$$

43. The BEST material for the core of an electromagnet is

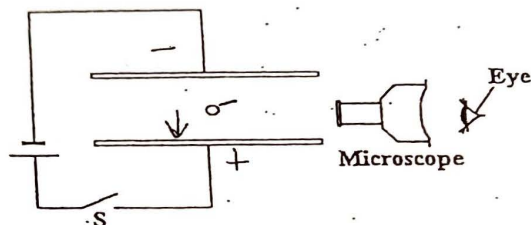
- (A) iron
(B) wood
(C) steel
(D) aluminium

A iron - magnetically soft

$\begin{array}{r} 275 \\ +150 \\ \hline 225 \end{array}$

- 12 -

Item 44 refers to the following diagram which shows a small negatively charged dust particle midway between two plates and observed with a microscope.



-ve
 attracted
 to
 the
 plate

44. Which of the following arrows correctly indicates the direction of the dust particle when the switch, S, is closed?

- (A) $\rightarrow$
 (B) $\leftarrow$
 (C) $\uparrow$
 (D) $\downarrow$

D

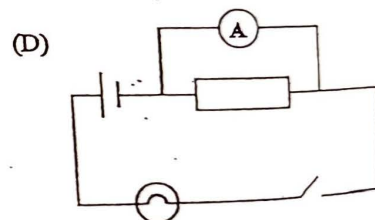
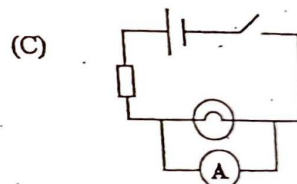
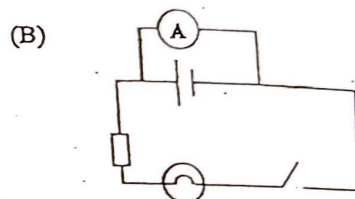
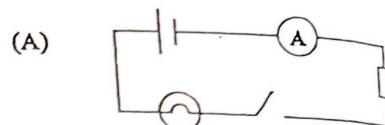
45. Which of the following relationships gives a correct value for the combined resistance of resistors R_1 , R_2 and R_3 connected in parallel?

- (A) $R_T = R_1 + R_2 + R_3$
 (B) $R_T = \frac{R_1 R_2 R_3}{R_1 + R_2 + R_3}$
 (C) $R_T = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
 (D) $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

D

46. Which of the following illustrates the MOST suitable circuit for measuring the current flowing through a lamp?

A



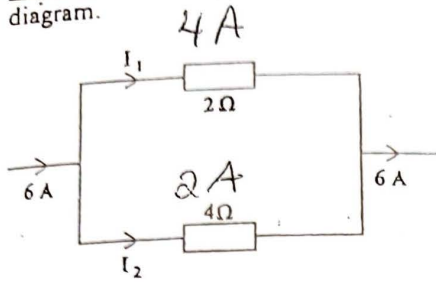
Ammeter
 must be in
 series

really
 t

R R
 $2:4$ Low R
 $I: 4:2$ Higher I

- 13 -

Item 47 refers to the following circuit diagram.

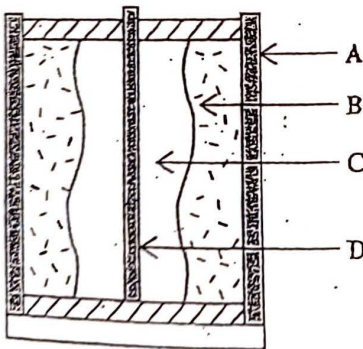


47. Using the information given in the circuit diagram, which pair of values of I_1 and I_2 is CORRECT?

- | | I_1 | I_2 |
|-----|-------|-------|
| (A) | 1 | 5 |
| (B) | 2 | 4 |
| (C) | 3 | 3 |
| (D) | 4 | 2 |

Current splits as a ratio of resistance

Item 48 refers to the following diagram of a dry cell.



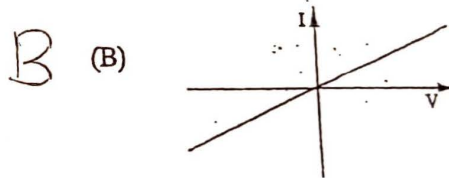
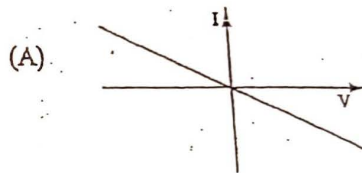
48. Which arrow points to the depolarizing agent of the cell?

- (A) A
 (B) B
 (C) C
 (D) D

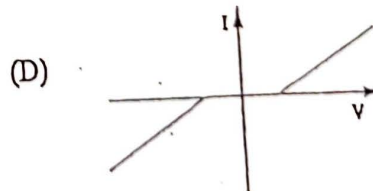
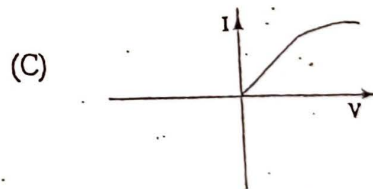
49. Which of the following expressions can be used to calculate the electrical power of a circuit?

- (A) $\frac{I^2}{R}$
 (B) IV
 (C) ItV
 (D) V^2R
- $P = IV$
 $P = I^2 R$
 $P = \frac{V^2}{R}$

50. Which of the following diagrams represents the current/potential difference relationship for a metallic conductor at a constant temperature?



Ohm's law



GO ON TO THE NEXT PAGE

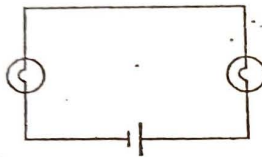
51. Which of the following quantities is the SMALLEST?

10-6

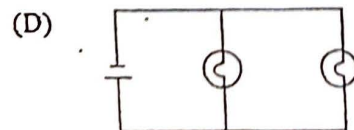
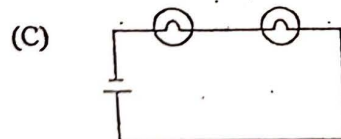
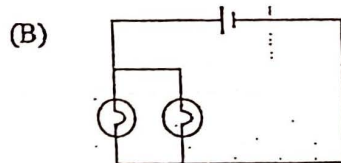
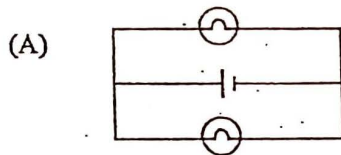
A

- (A) 1 microampere
- (B) 1 milliampere
- (C) 10^{-3} ampere
- (D) 10^3 ampere

Item 52 refers to the following diagram which shows an electrical circuit with a cell and two filament bulbs.



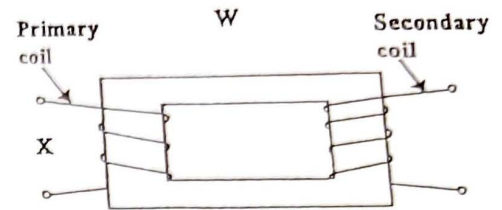
52. Which of the following circuits are electrically the same as the circuit above?



C

Two bulbs
in
series

Item 53 refers to the following diagram.



53. Appropriate labels for W and X would be

A
ac
input

- | | W | X |
|-----|-----------------------|------------|
| (A) | step-up transformer | a.c. input |
| (B) | step-up transformer | d.c. input |
| (C) | step-down transformer | a.c. input |
| (D) | step-down transformer | d.c. input |

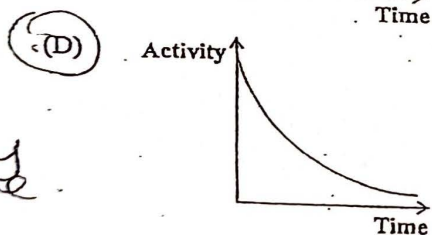
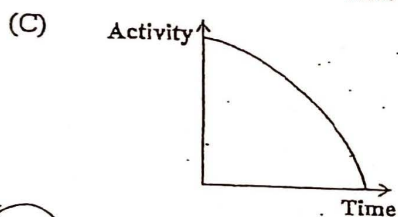
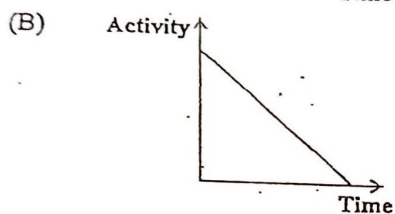
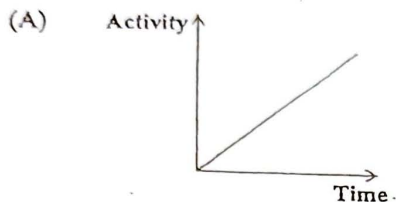
54. The electric power supplied to homes and factories is alternating current (a.c.) rather than direct current (d.c.) because

B

- (A) the use of a.c. reduces electrical hazards
- (B) the use of a.c. reduces transmission losses
- (C) a.c. enables a wider variety of appliances to be used
- (D) a.c. power lines are less likely to be struck by lightning than d.c. power lines

ac transmission
always has
less E losses

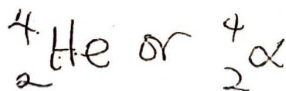
55. Which of the following graphs shows how the activity of a radioactive source varies with time?



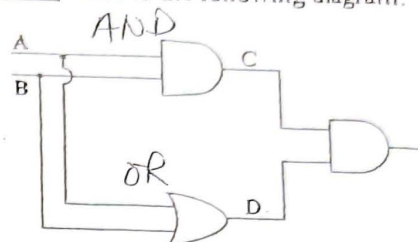
Exp
decay
curve

56. An alpha particle is

- (A) a helium nucleus
(B) a positive electron
(C) a high-speed electron
(D) high-frequency electromagnetic radiation



Item 57 refers to the following diagram.

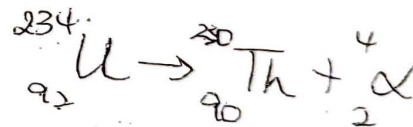


57. What is the output at C and D when a 0 0 input is made at A and B?

A	C	D
(A) 0	0	0
(B) 1	1	1
(C) 0	1	1
(D) 1	0	0

58. An isotope of uranium, ${}^{234}_{92}\text{U}$, changes to ${}^{230}_{90}\text{Th}$. This is an example of

- (A) fission
(B) fusion
(C) α decay
(D) β decay



Items 59–60 refer to the following information.

<u>Nucleus</u>	<u>Neutron Number</u>	<u>Proton Number</u>
P	126	82
Q	126	83
R	125	82
S	146	92

59. Which pair of atomic nuclei are isotopes?

- (A) P and Q
(B) Q and R
(C) P and R
(D) Q and S

same atomic
different
mass ~~\$~~

60. Which two elements could occupy adjacent positions in the periodic table?

- (A) P and Q
(B) P and R
(C) Q and S
(D) R and S